

"The Green Shift: Unveiling Europe's Path to Clean Energy"

CS910: StudentId-5588984

Abstract—Within the global movement towards sustainable energy, Europe is a key player in the field of renewable energy. This paper uses an extensive set of data covering the early 2000s to 2020 to lay out Europe's ongoing shift to renewable energy. Important variables like CO2 emissions, the transition from fossil fuels to renewable sources, trends in primary energy consumption and renewable electricity capacity with a focus on European nations are examined. The use of clustering model, which is intended to reveal underlying patterns and classifications within the data and provide an in-depth knowledge of Europe's energy landscape, is an essential component of this study. The study also makes use of various data visualization strategies to convey these complicated connections and trends. This helps to inform strategic and policy decisions related to the development of sustainable energy.

I. INTRODUCTION

In today's global scenario, switching to renewable energy is essential for sustainable development as well as the environment. In this fundamental change, Europe has emerged as a head, leading the way in the transition to more sustainable and clean energy sources with notable progress [8]. In that case, this study offers a comprehensive analysis of Europe's journey towards sustainable energy by exploring this transition, analysing the trends and developments from the early 2000s to 2020. This approach supports the identification of specific energy profiles among various regions in Europe, thereby providing valuable perspectives on their distinct challenges and progress in the integration of renewable energy. The study not only reflect the current state of renewable energy in Europe but also provide insights into the region's commitment to a green energy future by combining data analytics quality with cluster modelling. This study can be a useful resource for international energy transformation initiatives and is a reference for Europe's critical role in forming a sustainable future.

II. BACKGROUND

A. Overview of Renewable Energy

When it comes to fighting climate change and reducing reliance on fossil fuels, renewable energy—which includes resources like solar, wind, hydro, biomass, and geothermal energy—is at the centre of international efforts[9]. These energy sources are essential for making the shift to a low-carbon, sustainable future. They provide a way to reduce the negative effects that conventional energy sources have on the environment, like air pollution and greenhouse gas emissions.

B. Carbon Emissions and Low-Carbon Electricity

A critical component of the dataset is the information on CO2 emissions and the proportion of low-carbon electricity in the total electricity mix [5]. Carbon emissions are a key indicator of the environmental impact of energy production and consumption. The shift to low-carbon electricity, which includes renewable sources and potentially nuclear energy, is essential for reducing greenhouse gas emissions and combating climate change[10]. Understanding the trends in carbon emissions and the growth of low-carbon electricity is crucial for evaluating the effectiveness of renewable energy policies and the overall progress towards environmental sustainability goals.

C. The role of finance in the transition to renewable energy

This study examines how Europe's transition to renewable energy is impacted by changes in GDP growth, per capita income, energy prices, and international financial flows. GDP measures show financial potential for renewable energy investment, and shifts in fossil fuel prices show how they affect the competitiveness of renewable energy [2]. Most importantly, aid flows to developing nations highlight the part that foreign aid plays in global sustainability initiatives. Combining these factors provides a solid grasp of how local economic situations and outside funding interact to influence the renewable energy setting [1]. This method offers vital details into the hurdles and economic forces behind Europe's transition to sustainable energy practices.

D. Energy Efficiency and Renewable Integration

The relationship between energy intensity, per capita energy consumption, and the percentage of renewable energy in total energy consumption is examined in this analysis. Energy efficiency in relation to economic output is reflected in energy intensity (MJ/\$2011 PPP GDP), whereas primary energy consumption per capita reveals the levels of individual energy usage. By contrasting these with the percentage of renewable energy in total energy consumption, how well various nations are incorporating renewables into their energy mix can be learned[3]. Therefore, in connection with Europe's efforts to move towards a greener, more sustainable energy future, this comparison may highlight trends in how different countries strike a balance between economic growth and sustainability and energy efficiency.

III. LITERATURE SURVEY

This journal paper [4] examines the effect of natural gas, nuclear energy and renewable energy on the GDP and

carbon emissions across the world's top ten carbon-emitting countries from the years 1990 to 2014. The paper uses econometric models to understand the long-term effects and the relation between energy consumption with economic and environmental indicators. An interesting find is that the fluctuating contribution of various energy sources to the economic growth and carbon-dioxide reduction, nuclear and renewable energy has been very important for addressing the climate change and supporting economic development. But, the study has some limitations, including the potential model specificity and granular data is needed to understand the effects at the country level, which might implicate some policies. The paper covers a gap in research while focusing on the top emitting countries providing important insights.

The article talks about a panel analysis of data across Europe and Latin America [6]. It conducts a thorough analysis of renewable energy (RE) policy effectiveness in EU and Latin America using econometric methods. The study weighs-in the impact of various RE support policies in the given regions over a time of 2 decades, focusing on their capacity to trigger new investments in RE technologies. Key findings included in the paper the positive and significant effect of promotion policies in general on RE investments, the success of specific policy instruments such as feed-in-tariffs, renewable portfolio standards, auctions and the importance of economic and energy-related determinants in the RE diffusion. The paper highlights the varying effectiveness and steps taken to the RE policy between EU and Latin America, reflecting upon the differences in economics, political and energy sector dynamics.

The journal paper examines the integration of renewable energy in South America's transition from fossil-based fuels [7]. The paper mainly focuses on Ecuador while analyzing its energy system and proposing realistic green energy sources to meet the demands by the year 2050. The study highlights the major role of hydro, solar and wind power in changing Ecuador's energy matrix, while discussing the impact on production levels and the quality of life improvements, with the aim of achieving a 100% renewable electricity generation system by 2050.

IV. THE DATASETS

A combination of various datasets is utilized in this study for better understanding of the objective and analytical purposes. The datasets are as follows:

A. Global Data on Sustainable Energy: Focus on European Countries

This primary dataset, which was initially a global collection covering a broad range of nations, has been narrowly focused to include only European countries. A wide range of energy-related statistics from several European nations are contained in the dataset, which comprises the years 2000 to 2020. Key metrics, including the percentage of renewable energy in total consumption, electricity generated from renewable sources, CO₂ emissions per capita, energy

consumption per person, electricity production from renewables, and GDP growth indicators, are provided. It focuses on the shift towards renewable energy. This dataset is useful for assessing environmental impacts, tracking trends in the adoption of renewable energy across Europe, and directing energy policy decisions.

B. Renewables: Generation by Source - TWh

A breakdown of Europe's renewable energy production in 2021 and 2022 is given by this dataset, which includes output from solar, wind, and other renewable sources. It provides information on the spread and use of different renewable technologies by providing generation volumes expressed in terawatt-hours (TWh) and annual growth rates.

C. Renewables - Solar and Wind (Installed Capacity)

This dataset charts the installed capacities for solar and wind energy in Europe for a time period. It tracks annual growth rates and determines the overall capacity in each sector, especially from 2012 to 2022, and the share of solar and wind in the overall renewable capacity for the year 2022, offering insight into the acceleration of the adoption of renewable energy sources.

D. Fossil fuel: Coal-Focused Energy Datasets

Three separate subsets are included in this dataset: "Coal: Production - Exajoules," which tracks the extraction of coal across European nations starting in 1981; "Coal: Consumption - Exajoules," which describes the usage of coal in these countries since 1965; and "Coal: Trade Movements," which records the dynamics of import and export. When combined, these datasets offer a historical and current view of coal's place in Europe's energy system, changes in energy production and consumption trends, and the evolving role of coal in global trade as Europe moves more and more toward renewable energy sources.

E. Nuclear Energy: Generation and Consumption

This dataset traces Europe's nuclear energy output in terawatt-hours and its consumption in exajoules from 1965. It provides insights into the environmental implications of nuclear waste and evaluates Europe's progress towards greener, sustainable energy practices.

F. Impact of Gas Flaring on European Emissions

This dataset provides gas flaring volumes and the consequent CO₂ emissions since 1975, providing a critical measure of environmental impact and resource management effectiveness within Europe's energy sector.

V. HYPOTHESIS

This entire analysis is completely based on time series, to draw insights on how Europe has evolved over the years with respect to renewable energy could be analysed that Europe's path towards renewable revolution and greener environment is marching forward positively for decades. The following hypothesis have been drawn that supports this decision:

- • Decrease in electricity production from other sources of energy (fossil fuel and nuclear) could indicate the adaption of renewable energy by Europe.
- • Rapid adoption of renewable energy with integration of renewable sources like (solar, wind and hydro) could have increased Europe's energy mix over the years along with positive growth rate.
- • A higher percentage of low-carbon electricity and increased renewable energy generation can tend to reduce per capita CO2 emissions by a significant amount in all of the European countries
- • Energy intensity level of primary energy in Europe when dropped can potentially indicate a rise in energy efficiency in relation to economic output.
- • There could be a chance that the amount of financial aid given to developing nations in Europe for clean energy projects contributed significantly to the expansion of renewable energy capacity and renewable energy share.
- • Decreased trends in fossil fuel(coal) production and trading over the years could be an indicator of increased Renewable energy adaption.

VI. DATA CLEANING

A. Filtering European Data

Data that was originally global in scope was narrowed down to only include nations in Europe. This was accomplished by using the dataset's country identifiers, which ensured that any further analysis would be geographically relevant to Europe.

B. Time Frame Alignment

A common timeframe was established by balancing datasets from various time periods. This involved truncating datasets to match a standardized period, crucial for ensuring time frame consistency in the study and to enable comparative analysis across various energy sectors.

C. Combining datasets

Database join operations and data alignment techniques were used to integrate the "Coal-Consumption," "Coal-Production," and "Coal-Trade Marks" datasets. In order to match records across the datasets based on common keys like country names and years, SQL join queries were used in a relational database environment during this process. By using this technique, the unified dataset's records were guaranteed to accurately represent corresponding production, consumption, and trade data for the same time frame and regions in Europe.

D. Handling Missing Values

Missing values were handled using the averages and patterns of the surrounding data. To preserve the integrity of the entire dataset in cases where estimation was not efficient, default values were assigned.

E. Handling outliers

The Z-score method was used to identify data points that showed extreme variance, or deviations from the mean of more than three standard deviations. Furthermore, data distribution was examined in more detail using the Interquartile Range (IQR) approach, particularly for the "Coal Consumption" and "Natural Gas Flaring" datasets. To preserve data integrity, outliers found using these techniques were substituted with the mean or median of the corresponding attributes. In the case of the "Natural Gas Flaring" and "CO2 Emissions from Flaring" datasets, extreme outliers representing sudden rises in emissions or flaring activities were carefully examined and corrected.

VII. DATA PROCESSING

First, the relationship between the attributes is analysed to draw meaningful insights that can contribute to the objective of this study before stepping into feature engineering.

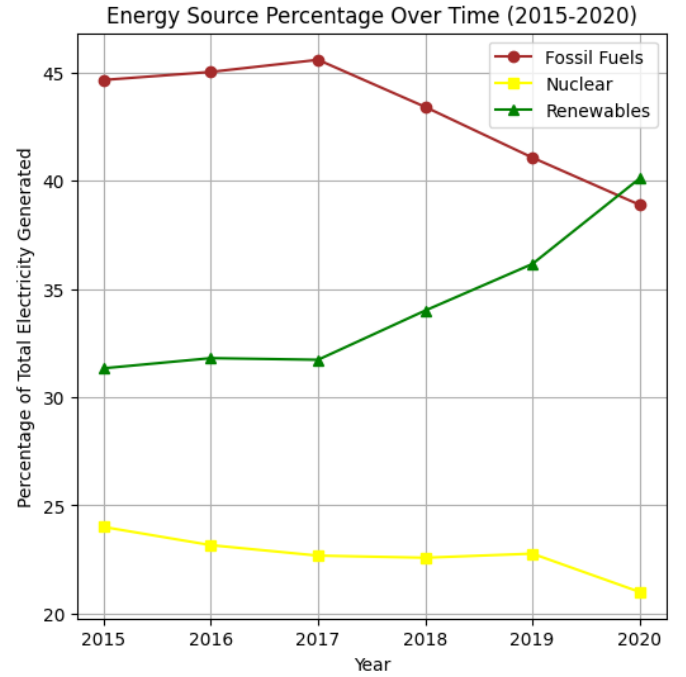


Fig. 1. Percentage of Electricity produced form Energy Sources Percentage like Fossil fuels, Nuclear and Renewables from 2015 to 2020.

From Figure 1, it is evident that reliance on fossil fuels for electricity shows a consistent decline. On the contrast, contribution of Renewables is peaked with continuous increase. It also depicts an intersection between Fossil fuel and Renewables showcasing that by 2020, Renewables have taken over Fossil fuel with respect to electricity production.

On the left, it is clear that of all countries, United Kingdom has a significantly higher total renewable energy generation compared to other countries, with major contribution coming from wind energy. On the right, France leads in growth rates, particularly in solar energy. Figure 2 suggests that there could be recent investments or advancements in solar technology. Germany follows closely, with a balanced growth rates of

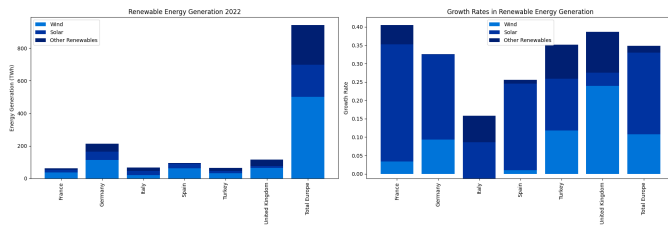


Fig. 2. Recent generation of energy from renewables like wind, solar and other renewables vs the growth rate in renewable energy generation in 2022

renewables which could indicate a consistent approach to expanding the renewable energy infrastructure. The analysis on whole implies that the diverse contribution of wind, solar, and other renewables across different countries showcases Europe's extensive approach to increasing sustainable energy production.

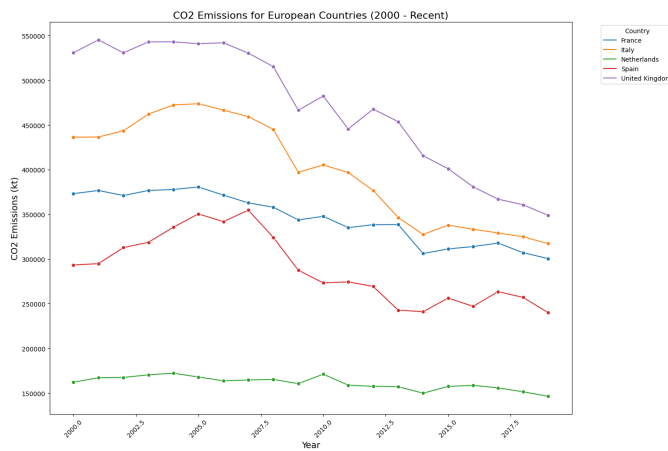


Fig. 3. Carbon dioxide emissions from European countries from 2000 to 2020

Figure 3 line graph depicts a trend where CO2 emission is gradually decreased from 2000 to recent years for several European countries. This show Europe's consistency towards adapting sustainable practices. The analysis showcases how countries like UK and France have reduced CO2 emission significantly when compared to other. Meanwhile. Netherlands shows less significance but still has a consistent decline, contributing to this overall reduction trend. The analysis on whole, captures the environmental action taken by Europe over the past two decades.

Figure 4 depicts two upward tracks in renewable energy share and financial aid to developing countries within main countries over two decades. It reflects an extensive investment in the renewable sector by Europe that correlates with the rise in percentage of energy produced from renewable sources. Overall, the trends indicate a strategic improvement of renewable energy capacities that are driven by environmental priorities and international aid by the European countries.

Figure 5 is to understand Europe's position in producing coal on a global level. The graphs show that from 1981 until recently, Europe has been producing less coal. Compared

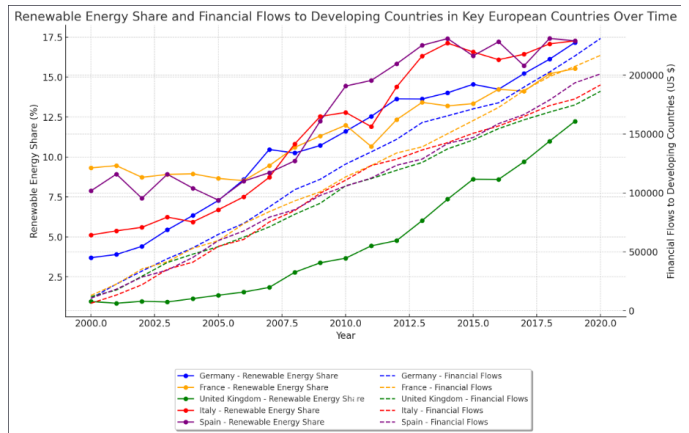


Fig. 4. Renewable Energy share and financial flows to developing countries over time

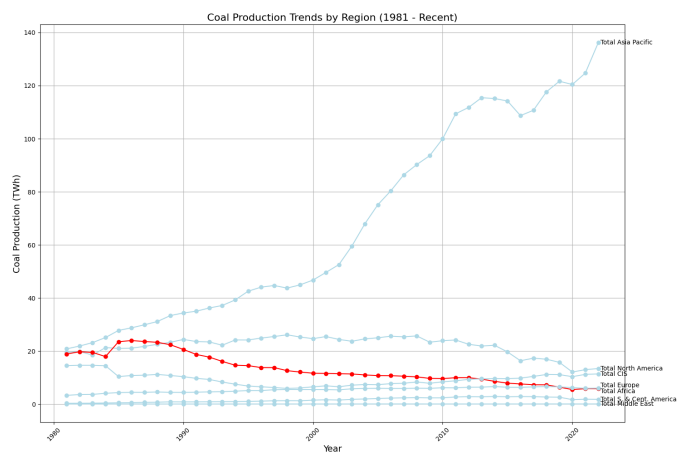


Fig. 5. Coal production trend analysis by major continents from 1981 to recent years

to other continents, initially, Europe has showed a decent amount of production from which it was gradually decreased. By producing less coal, Europe is working towards the goal to not harm the environment and use energy that won't run out.

Figure 6 shows the amount of natural gas flaring where it depicts a high volume of natural gas flaring in early years of 1980's. This could be an easy example of downward trend in which after 1980, even though the flaring shows complicated waves with some ups and downs, it showcases a gradual decrease eventually.

Energy intensity is the amount of energy used per unit of GDP and a lower number implies that the country's economy is using energy more efficiency's analysis shows a declining trend in energy intensity over time. Therefore, Figure 7 that countries are becoming more efficient in their use of energy. This type of decrease tends to show improvements in energy efficiency and a shift towards less energy-intensive industries.

Analysing the correlation between the attributes:

Figure 8 shows strong positive correlation between 'Electricity from renewables (TWh)' and 'Low-carbon electricity (% electricity)', which indicates that as European countries

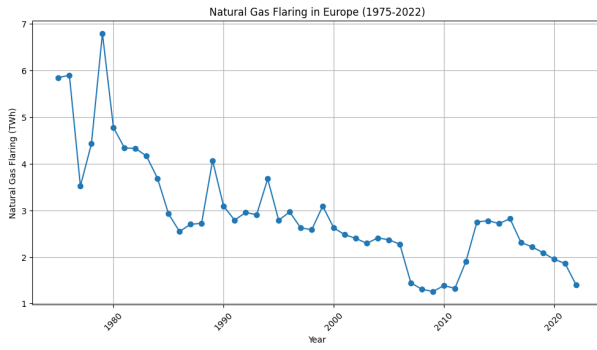


Fig. 6. Natural gas Flaring in Europe from 1975 to 2022

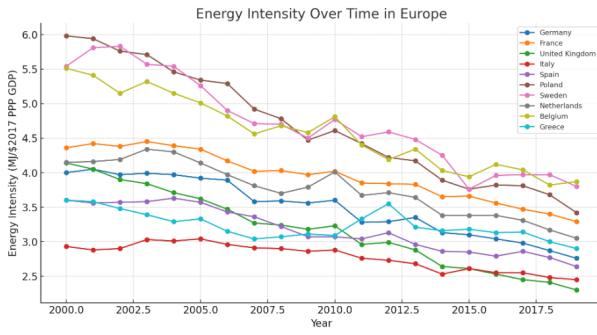


Fig. 7. Energy use per unit of GDP at purchasing power parity over time in Europe

generate more electricity from renewable sources, the proportion of low-carbon electricity also increases. Additionally, 'Renewables (% equivalent primary energy)' is highly correlated with 'Renewable energy share in the total final energy consumption', suggesting that as the primary energy from renewables rises, their share in total energy consumption also increases. From this analysis, it is found that these relationships could be the main indicators that would contribute to Europe's shift to renewable energy.

A. Feature Engineering

Fast Fourier Transform (FFT) is a technique in mathematics that converts a time-domain signal into its constituent frequency components to reveal the different periodic patterns and frequencies present in the data. This plays a major role in time-series analysis since it is equipped to analyse time-based datasets efficiently.

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-\frac{i2\pi}{N}kn} \quad (1)$$

- $X(k)$ - represents the complex frequency component at index
- $x(n)$ is the input signal at time index n
- N is the total number of data points in the time-domain signal
- k ranges from 0 to $N-1$ representing different frequency number

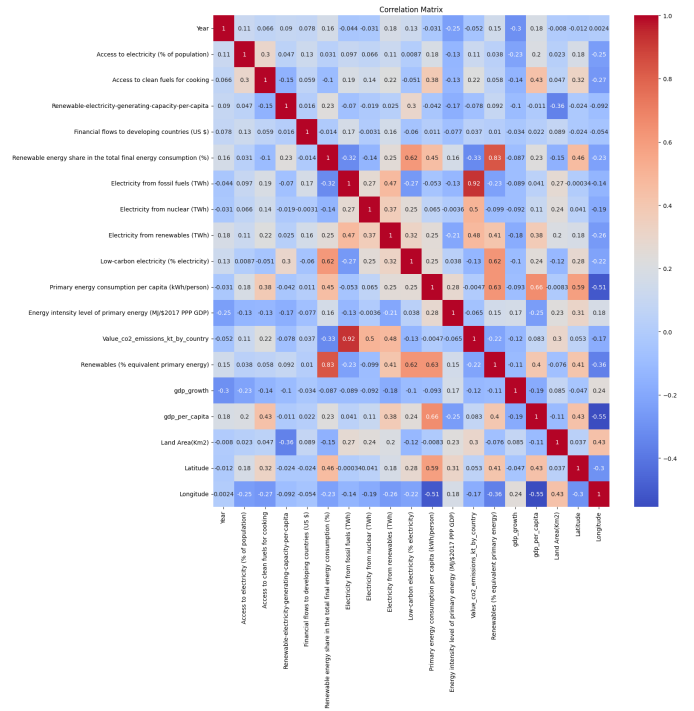


Fig. 8. Correlation matrix indicating correlation values between the attributes

From the results of visual analysis and correlation matrix, "Renewable energy share in the total final energy consumption (%)" is considered to be the suitable feature for processing through FFT.

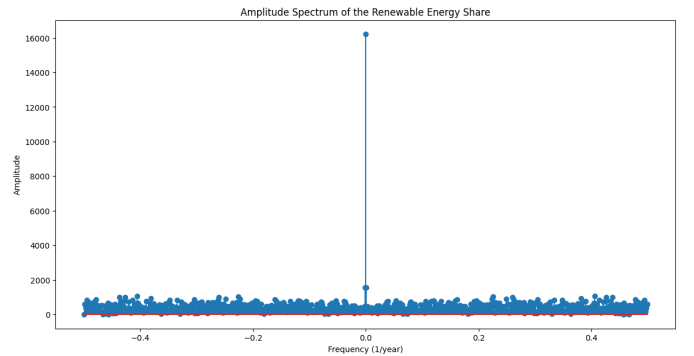


Fig. 9. Amplitude spectrum

The peak at the zero frequency indicates that the most suitable component is a constant, non-repeating variable from which it can be suggested that a stable average level of renewable energy share is analysed throughout the time period. The points represent less significant variations when they are compared to the overall level.

B. Building a Prediction Model

To align with the objective of the study which implies that Europe is marching towards renewable energy adoption, the renewable energy share for near future is forecasted to see

if the renewable energy share value increases in the future also.

1) *Linear Regression*: The predictor variables stored in the 'X' excluding highest amplitude, highest frequency, and Entity whereas the target variable- "Highest amplitude" in the 'Y'. MSE is a measure of the average squared difference between the predicted values (from the regression model) and the actual values (observed in the test dataset).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

Smaller MSE values indicate a better fit, with zero MSE representing a perfect match between predictions and actuals. Here, MSE value is 0.0 indicating perfect predictions.

R squared quantifies the proportion of variance in the target variable that is explained by the predictors in the regression model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

Here, R2 is 1.0, signifying that the model effectively explains all variance in the 'highest amplitude' variable.

2) *K-means Clustering*: Here, K-Means is used because, in multidimensional data such as the standardized economic and energy-related features provided, it is especially well-suited for detecting independent, spherical groups. Figure 10

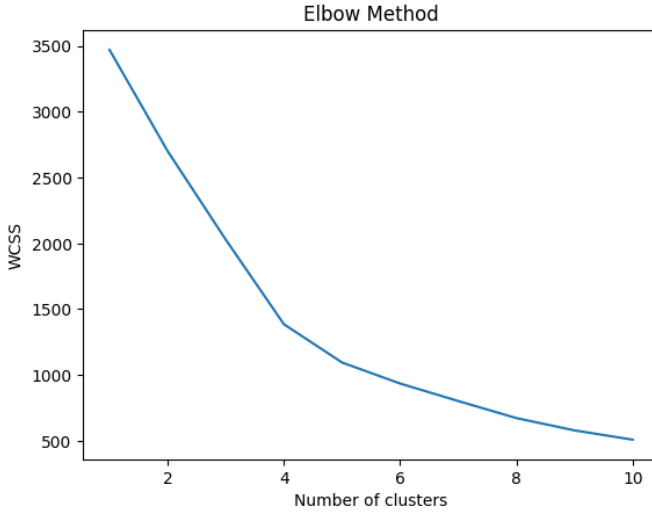


Fig. 10. Elbow Method by K-Means clustering

illustrates the Elbow Method used for finding the optimal number of clusters for K-Means clustering. The features include access to electricity, renewable energy usage, CO2 emissions, and GDP per capita, which have been normalized for equal impact in the analysis. For the given variables, this optimal point can be seen between 3 and 4 clusters which indicates a balance between model simplicity and clustering accuracy.

- **Silhouette Score**: The Silhouette Score is a metric used to test the quality of clusters created by an algorithm

(here K-Means). A Silhouette Score of 0.75 indicates a high level of separation between clusters. It suggests that, on average, data points are much closer to their own cluster centre than to the centres of other clusters. this implies that the clusters are well-defined.

VIII. FINAL PREDICTION

The final prediction is done with highest amplitude in 'Y' and results in forecasting the renewable energy share in a positive trend.

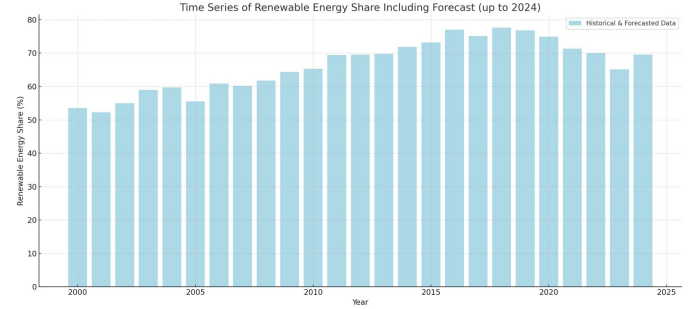


Fig. 11. Forecast of Renewable energy share till 2024

From Figure 11, it can be implied that Europe is indeed in its path towards a reaching the renewable revolution and sustainable environment.

IX. ANALYSIS

The main objective of the study is to show Europe is showing positive response in increasing the renewable energy production and share in overall energy mix. Various visual data analysis is performed to gain information on how the data depicts a relationship among themselves for showing Europe's adoption towards renewable energy. From the correlation matrix features having strong relationship are identified and selected for feature engineering. In Feature Engineering, From all the previous analysis "Renewable energy share in the total final energy consumption (%)" attribute is processed through the Fast Fourier Transform Technique from which the highest amplitude is calculated. Predictive model is build using linear regression with MSE value (mean squared error) 0.0 and representing a perfect match between the prediction and the actual value. The R^2 value is 1.0 which signifies that the model effectively explains all the variance in the 'highest amplitude' variable. The linear regression model is followed by K-means clustering. The Elbow method is then used to find the optimal number of clusters for the K-means clustering, the optimal values are then found out to be 3 and 4. Silhouette score of 0.75 indicates that the clusters are well defined. The renewable energy share in a positive trend is predicted using the target value as the highest amplitude in the final prediction. Thus, it is proved that Europe is indeed a leading renewable energy consumer in comparison to other countries.

X. CONCLUSION

It is clear from Europe's journey towards green energy from the early 2000s to 2020, marks an important paradigm shift in global energy practices. Through the study done, it is revealed that a commendable reduction in fossil fuel dependency which is replaced with a robust growth in renewable energy sources like solar and wind energy, paving the path to a considerable decrease in carbon-di-oxide emissions. This transition does not only reflect Europe's commitment to environmental leading but also sets a standard for global sustainability efforts. But the journey is has its own challenges. The disparities in renewable energy adoption across various European regions, factored in by economic, technological and policy variables are addressed. These hurdles emphasise the need of nuanced, region-specific approaches in advancing the green energy goals. From a policy perspective, the findings give vital insights for decision-makers. It advocates for policies that balance economic growth with environmental sustainability, emphasizing the need for innovative financial models and advancements in technology to foster more green energy adoption. This also suggests a more integrated European energy policy framework, capable of leveraging the collective strengths and addressing shared challenges. Looking ahead, promising areas for future research are identified, including exploring the emerging renewable energy and assess the socio-economic impacts in the energy transition and confirming the role of international collaboration in the global sustainment of energy. The criticality of Europe's continued commitment to renewable energy affirms not just as a regional priority but acts as a cornerstone for global efforts to battle the climate change and promote a sustainable future. The commitment requires persistent innovative and strategic policymaking and collaborative efforts to ensure that the gains made in the last twenty years not only sustained but further amplified in the coming years.

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